JOEY HUERTAS

Computer Engineering Graduate | Aspiring Software & Hardware Engineer

Pampanga, Philippines | joeyhuertas12@gmail.com | +63 938 692 9292
LinkedIn: linkedin.com/in/joey-huertas-8743b7417 | GitHub: github.com/huertasjoey | Portfolio: joey-port-folio.vercel.app

SUMMARY

Computer Engineering graduate who builds across hardware and software, from microcontrollers and PCBs to full-stack web apps and AI vision. Architected and built the hardware of LycoDry, an AI tomato grading and drying machine. Detail-oriented, quick to learn, and seeking an entry-level role in software or hardware engineering.

EDUCATION

Bachelor of Science in Computer Engineering

Pampanga State University (formerly Don Honorio Ventura State University)
Class of 2026

- Focus areas: embedded systems, digital logic, circuit and PCB design, computer vision, programming

EXPERIENCE

Computer Engineering Intern (OJT Trainee)

AI/ML Center, Pampanga State University
February 2026 – May 2026

- Installed, maintained and troubleshot computer hardware for the center.
- Installed and configured software and carried out routine system maintenance.
- Diagnosed and resolved hardware and software issues reported by users.
- Worked with the team to complete assigned technical tasks on schedule.

THESIS PROJECT

LycoDry: An Automated Solanum Lycopersicum Quality Grading and Dehydration System

Project Architect & Hardware Engineer, team of five
June 2026 · YOLOv11, Raspberry Pi 5, ESP32, Python, C++

- Architected the full system and designed and built all of its hardware: camera station, servo sorting, pneumatic slicer, conveyor and drying chamber.
- Integrated a YOLOv11 model (0.876 mAP@0.5) that sorts tomatoes into Market, Processing and Reject grades with 82–93% accuracy.
- Dried tomatoes in 24 hours versus 8 days of sun drying; the DOST lab test found no Salmonella, while sun-dried samples tested positive.
- Tested, debugged and refined the machine with a farm owner, who rated its usability 4.55 / 5.00.

OTHER PROJECTS

- **Kaizen** (React, Supabase, PostgreSQL): habit tracker, planner and private diary web app with user accounts. Live site
- **Fieldnote** (React, Express, MongoDB): magazine-style blog with an admin area and secure, session-based login. Live site
- **Aquarium Monitor** (KiCad, ATmega328P, ESP-01): Wi-Fi board, designed and hand-etched, that reads pH, TDS and temperature and runs fans and a pump.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, JavaScript, TypeScript
- **Web Development:** React, HTML, CSS, Tailwind CSS, Node.js, Express.js
- **Databases & Backend:** Supabase (PostgreSQL), MongoDB, MySQL, REST APIs
- **Embedded Systems & IoT:** ESP32, Raspberry Pi 5, Arduino, ATmega328P, sensors, servo motors
- **AI & Computer Vision:** YOLOv11 object detection, dataset preparation and model training
- **PCB Design & Hardware:** KiCad (schematic, PCB layout and routing), etching and soldering, computer assembly and troubleshooting
- **Tools:** Visual Studio Code, Git, GitHub
- **Soft Skills:** problem-solving, teamwork, communication, adaptability, time management